Inter Quartile Range

<u>Key Discussion Point:</u>

- **<u>IQR:</u>**
  Q3-Q1
- **<u>Lower Range of outlier:</u>**
  Cannot be less than Q1- (1.5)(IQR)
- **<u>Higher Range of Outlier:</u>**
  Cannot be greater than Q3 + (1.5)(IQR)
- **<u>Significance of 1.5</u>**

1. **Interquartile Range (IQR)** represents the middle 50% of your data (the crux of the distribution).

- **Q1:** The 25th percentile.
- **Q3:** The 75th percentile.
- **IQR = Q3 - Q1**

- By using **1.5 times IQR** as a buffer, we essentially mean that we can allow a value to be a certain distance away from the middle, but if it wanders too far, it can no longer be part of or representative of the group.
- The number **1.5** wasn't a random choice. It was popularized by John Tukey, a famous statistician.

- If for argument and smaller number say “1.0” is chosen, we could flag too many points as outliers even if they were just slightly unusual.

- In a **Normal(Gaussian) Distribution** , the range between Q1 - 1.5(IQR) and Q3 + 1.5(IQR) covers approximately **99.3%** of all data.

Appendix

Gaussian (The "How Likely?")

Gaussian (often called a **Normal Distribution**) describes how data is spread out. It is the famous "Bell Curve" where most observations cluster around a central average, and probabilities taper off equally in both directions.